F.E. , Div. E

(Artificial Intelligence & Data Science)

Welcome to

<u>Thakur Shree DPS College of Engineering & Management</u>

INSTITUTE VISION

"To be a leader in technical education, producing globally competent and socially responsible professionals, who will contribute to technological and societal advancements through innovation and ethical practices."

INSTITUTE MISSION

1. Deliver Comprehensive Technical Education

To provide a well-rounded education that integrates technical expertise, practical skills, and creativity, empowering students to solve complex problems.

2. Promote Research and Innovation

To foster a culture of research, creativity, and innovation to generate new knowledge and develop sustainable technologies that address local and global challenges.

3. Develop Ethical and Responsible Engineers

To instill ethical values and social responsibility in students, emphasizing sustainability, diversity, and the positive impact of technology on society.

4. Encourage Lifelong Learning and Professional Growth

To encourage continuous learning, leadership development, and effective communication to adapt to the changing technological landscape and foster holistic personal growth.

DEPARTMENT VISION

"To nurture an outcome-based and innovative ecosystem that shapes globally competent, ethical, skilled, and visionary professionals in AI and Data Science to serve society, drive innovation, and advance technology."

DEPARTMENT MISSION

Mission 1: Deliver Industry-Focused and Interdisciplinary Education

To provide practical, outcome-driven learning that blends AI, Data Science, and multidisciplinary knowledge to solve real-world challenges.

Mission 2: Inspire Research and Innovation

To apply modern tools and engage individuals in research activities for the technological advancements to nurture the culture of research and startups.

Mission 3: Nurture Ethical and Global Professionals

To imbibe ethical values, social responsibility, and cultural awareness to prepare professionals for a diverse, global AI platform.

Mission 4: Support Lifelong Learning and Holistic Development

To encourage lifelong learning, leadership, teamwork, and communication for sustained holistic and professional growth.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

PEO-1: Professional Excellence

To apply analytical, problem-solving, and programming skills to solve complex challenges in Artificial Intelligence and related fields as skilled professionals.

PEO-2: Research, Higher Studies, and Innovation

To contribute to advancements in AI through research, innovation and higher studies while engaging in lifelong learning.

PEO-3: Innovative and Multidisciplinary Engagement

To demonstrate innovation, teamwork, communication, and an entrepreneurial mindset to contribute across multidisciplinary domains as competent professionals.

PEO-4: Ethical and Social Responsibility

To practice the profession ethically and responsibly to promote sustainable development and societal well-being.

PROGRAM OUTCOMES (POS)

PO1: Engineering Knowledge

Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solutions of complex engineering problems.

PO2: Problem Analysis

Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions

Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems

Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)

PROGRAM OUTCOMES (POS)

PO5: Engineering Tool Usage

Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World

Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, & WK7)

PO7: Ethics

Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work

Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams. (WK9)

PROGRAM OUTCOMES (POS)

PO9: Communication

Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance

Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-long Learning

Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOS)

ON SUCCESSFUL COMPLETION, GRADUATES OF B.E. (ARTIFICIAL INTELLIGENCE & DATA SCIENCE) DEGREE PROGRAM WILL BE ABLE TO:

PSO-1: Design and implement intelligent models using AI, Data Analytics, and statistical techniques to solve domain-specific problems using structured and unstructured data.

PSO-2: Apply data engineering tools, programming languages, cloud platforms and deployment strategies to build scalable, efficient, and real-time AI/ML implementations.

PSO-3: Apply engineering, science, and mathematical principles to design, develop, and implement effective solutions for complex problems, addressing environmental and societal needs.

SEMESTER PATTERN

SEMESTER I

SEMESTER II

SEMESTER- I SUBJECTS

- Applied **Mathematics – I**
- Applied **Physics**
- Applied **Chemistry**
- Engineering **Mechanics**
- Basic **Electrical &Electronics** Engineering
- Professional **Communication &** Ethics
- C **Programming**
- **Workshop** Practice

SEMESTER- II SUBJECTS

- Applied **Mathematics – II**
- Elective **Physics**
- Elective **Chemistry**
- Engineering **Graphics**
- **Python** Programming
- **Core Course** Subjects
- **Indian Knowledge** Systems
- **Workshop** Practice

ATTENDANCE NORMS

- ❑ **Minimum 75% attendance** in each subject in each subject (in each term)

- ❑ **Defaulter List** prepared monthly → Shared with all (Students & Parents)

- ❑ **Poor attendance** → May lose exam eligibility
  (*as per University of Mumbai guidelines*)

RULES FOR PASSING & PROMOTION TO S.E.

- **Minimum 40%** marks in each subject (Theory, Practical)
- **Must clear** set of Theory (Internal & University) &/or Practical (Termwork & Oral/ Practical) exams
- **Minimum 32 credit (Maximum 45** (23+22)) to be earned (SEM-I + SEM-II)
- Result of Previous Academic Year ≈ 80%

NEP 2020 WITH EFFECT FROM ACADEMIC YEAR (2024 - 2025) SEMESTER- I

Course Code	Course Description	Teaching Scheme (Contact Hours)			Credit Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	Practical	Total Credits
BSC101	Applied Mathematics-I	2	--	1	2	1	--	3
BSC102	Applied Physics	2	--	-	2	-	--	2
BSC103	Applied Chemistry	2	-	-	2	-	-	2
ESC101	Engineering Mechanics	2	-	-	2	-	-	2
ESC102	Basic Electrical & Electronics Engineering	3	--	-	3	-	--	3
BSL101	Applied Physics Lab	-	1	-	-	-	0.5	0.5
BSL102	Applied Chemistry Lab	-	1	-	-	-	0.5	0.5
ESL101	Engineering Mechanics Lab	-	2	-	-	-	1	1
ESL102	Basic Electrical & Electronics Engineering Lab	--	2	-	--	-	1	1
AEC101	Professional and Communication Ethics	2	--	-	2	-	--	2

NEP 2020 WITH EFFECT FROM ACADEMIC YEAR (2024 - 2025) SEMESTER- I

AEL101	Professional and Communication Ethics	--	2		--	--	1	1
VSEC101	Engineering Workshop-I	-	2	-	-	-	1	1
VSEC102	C Programming	-	2*+2	-	-	-	2	2
CC101	Induction cum Universal Human Values	2#	-	-	2	-	--	2
Total		**15**	**14**	**1**	**15**	**01**	**07**	**23**

* Two hours of practical class to be conducted for full class as demo/discussion.
# Course evaluation is activity-based which may be an individual or group of four students.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

CREDIT ASSIGNMENT: SAMPLE CALCULATION I

		THEORY CREDITS				TUTORIAL CREDITS	CREDITS
AM-I	IAT	ESE	2 Credits	+	TW	1 Credit	3 Credits
	~~IAT~~	ESE	0 Credits	+	TW	1 Credit	1 Credit
	IAT	~~ESE~~	0 Credits	+	TW	1 Credit	1 Credit
	IAT	ESE	2 Credits	+	~~TW~~	0 Credits	2 Credits
	~~IAT~~	~~ESE~~	0 Credits	+	~~TW~~	0 Credits	0 Credits

CREDIT ASSIGNMENT: SAMPLE CALCULATION II

EM & BEEE

	THEORY CREDITS				TUTORIAL CREDITS			CREDITS
IAT	ESE	2 Credits	+	TW	OR	1 Credit		3 Credits
~~IAT~~	ESE	0 Credits	+	TW	OR	1 Credit		1 Credit
IAT	~~ESE~~	0 Credits	+	TW	OR	1 Credit		1 Credit
IAT	ESE	2 Credits	+	~~TW~~	OR	0 Credits		2 Credits
~~IAT~~	~~ESE~~	0 Credits	+	TW	~~OR~~	0 Credits		0 Credits

Examination Scheme: Semester- I

Course Code	CourseDescription	Examinationscheme							
		Internal Assessment Test (IAT)			End Sem. Exam Marks	End Sem. Exam Duration (Hrs)	Term Work (Tw)	Oral & Pract.	Total
		IAT-I	IAT-II	Total (IAT-I) + IAT-II)					
BSC101	AppliedMathematics-I	20	20	40	60	02	25	--	125
BSC102	AppliedPhysics	15	15	30	45	1.5	--	--	75
BSC103	Applied Chemistry	15	15	30	45	1.5	--	--	75
ESC101	EngineeringMechanics	20	20	40	60	02	--	--	100
ESC102	BasicElectrical&Electronics Engineering	20	20	40	60	02	--	--	100
BSL101	AppliedPhysics Lab	--	--	--	--	--	25	--	25
BSL102	Applied Chemistry Lab	--	--	--	--	--	25	--	25
ESL101	EngineeringMechanics Lab	--	--	--	--	--	25	25	50
ESL102	BasicElectrical&Electronics Engineering Lab	--	--	--	--	--	25	25	50
AEC101	ProfessionalandCommunicationEthics	15	15	30	45	1.5	--	--	75
AEL101	ProfessionalandCommunicationEthics	--	--	--	--	--	25	--	25
VSEC101	Engineering Workshop-I	--	--	--	--	--	25	--	25
VSEC102	C Programming	--	--	--	--	--	25	25	50
CC101	Inductioncum Universal HumanValues	--	--	--	--	--	-	--	-
Total		**105**	**105**	**210**	**315**	**10.5**	**200**	**75**	**800**

EXAMINATIONS (CONTINUOUS ASSESSMENTS)

TOTAL: 23 | 22

- Internal Assessments: I & II
- Term Work & Practical/Oral
- End-Semester Exams

Grading system:

CGPI [10]

Academic Arrangements

- **Semester I:** 31 Aug. – Dec. 2026 | **Semester II:** 4 Jan. – May 2027
- **Internal tests:** Oct. & Nov. 2026 | Feb. & April 2027
- **End Semester Exams:** Dec. 2026 | May 2027
- **Results** declared ≈ 45 days after exams

PARENT / GUARDIAN COMMUNICATION

- **Defaulter lists & Exam results** will be shared on WhatsApp groups
- **Parent Teacher Meeting**
  - **Via. Class-In-Charge and/or HOD**

LIBRARY SERVICES

- **Book Bank** facility for textbooks
- Access to **Central Library & Reference Books**
- **Digital Library:** IEEE, NPTEL, E-Magazines, etc.
- Study areas & **reading halls** available

SOME USEFUL INFORMATION

- F.E. classrooms: 301 to 307
- F.E. Laboratories: I, II & III floor
- Library: 209
- F.E. Staffroom: 207
- College hours: 8 pm to 4 pm **or** 9 am to 5 pm
- Canteen: Ground floor
- Main Events: Freshers, Naad, Personality Development etc.

BEYOND THE CLASSROOM

- 💻 Technical Festivals
- 🧑‍💻 Hackathons & Innovation Activities
- 🎭 Cultural Events
- 🏆 Sports & Competitions
- 🤝 NSS – National Service Scheme
- 🌱 DLLE – Development of Learners and Learning Environment

Why participate?

Build confidence • teamwork • leadership • creativity • communication skills

MENTORING & STUDENT SUPPORT

- **Students will be assigned faculty mentors in groups**
- **Mentors will help students with:**
  1. Academic guidance
  2. General concerns and difficulties
  3. Personal and professional development
- *Your mentor is your first point of contact whenever you need guidance or support*
- Student leadership & responsibilities
  - Class and lab. representatives

TRANSPORT & HOSTEL SERVICES

- ➢ 🚌 College Bus Facility
- ➢ 🛺 Share Auto-Rickshaw connectivity
- ➢ 🏠 Hostel facilities are available for students

For detailed information regarding transport, hostel and other administrative matters, please contact the college administration.

SUMMARY

- **Well-structured** semester-based curriculum
- **Balanced mix of** theory, practical & projects
- Focus on **attendance, discipline & continuous assessment**
- **Parents & faculty** work together for student success
- Co & Extra Curricular activities

THANK YOU!

"Engineering is not only study of 45 credits / subjects but it is moral studies of intellectual life & the art of balancing!"